

Techfest 2026-27



इलेक्ट्रॉनिकी एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
**ELECTRONICS AND
INFORMATION TECHNOLOGY**



CycloProp: Advanced UAV Propulsion Challenge

Overview:

PUSHPAK Grand Challenge 2026: Towards Drone Excellence

A National Innovation Challenge under the PUSHPAK: National Mission on Drone Technology.

Funded by the **Ministry of Electronics and Information Technology (MeitY), Government of India.**

Hosted by **IIT Bombay.**

In Collaboration

With Drone Centre, Department of Aerospace Engineering

India is entering a new era of autonomous aerial systems, intelligent mobility, and drone-enabled digital infrastructure. The PUSHPAK Grand Challenge 2026 is a national platform created to identify and accelerate high-impact engineering solutions that can address real-world problems through drones, AI, robotics, sensing, communication systems, and autonomous technologies.

This is not a conventional hackathon or idea competition. The Grand Challenge is designed to take promising technical concepts through evaluation, mentoring,

prototyping, testing, and deployment-oriented development. Selected teams will work with researchers, domain experts, and industry mentors to transform innovative ideas into functional, field-ready prototypes.

Grand Challenge 2, presented by Drone Centre at IIT Bombay Techfest 2026-27, is titled "**CycloProp: Advanced UAV Propulsion Challenge**"

Guiding Professors / Mentors

Grand Challenge 2 (CycloProp: Advanced UAV Propulsion Challenge) will be guided by the following faculty mentors from the collaborating institutes:



Prof. Arnab Maity

IIT Bombay – Aerospace Engineering

Prof. Arnab Maity is an accomplished aerospace researcher and faculty member at IIT Bombay with strong expertise in dynamics, guidance, navigation and control of aerospace and autonomous systems. His key strength lies in combining rigorous theoretical foundations in control, estimation, optimization and system dynamics with their application to complex real-world aerospace problems. His research encompasses autonomous flight systems, UAV and drone technologies, guidance and navigation algorithms, adaptive and optimal control, and aerospace vehicle and propulsion-system control. With a strong analytical and systems-oriented approach, Prof. Maity contributes to the development of robust, intelligent and reliable autonomous platforms, particularly for applications where safety, precision and real-time decision-making are critical. His expertise makes him well positioned to contribute to advanced drone technology initiatives involving autonomous navigation, flight control, collision avoidance, multi-agent systems and next-generation UAV applications, while bridging fundamental research with technology development and mission-oriented outcomes.



Dr. Dhwanil Shukla

IIT Bombay – Department of Aerospace Engineering

Dr. Dhwanil Shukla is an aerospace engineering researcher and faculty member in the Department of Aerospace Engineering at the Indian Institute of Technology Bombay (IIT Bombay). His work focuses on aerodynamics, particularly rotorcraft design and aerodynamics, low-speed aerodynamics, and experimental flow-diagnostic techniques. He earned his B.Tech. in Mechanical Engineering from IIT Gandhinagar and his Ph.D. in Aerospace Engineering from Georgia Institute of Technology. Dr. Shukla is a two-time recipient of the Vertical Flight Foundation Scholarship and received the President's Gold Medal at IIT Gandhinagar. Through his teaching and research, he contributes to the development of experimental aerodynamics and emerging rotorcraft and aerial-vehicle technologies.

What Happens After the Grand Finale?

The journey does not end with the finale.

Prototype Development Support (Build & Test Phase)

Selected winning teams (upto 5 teams) will be invited to participate in a dedicated Prototype Development Support Programme, beginning December 2026 onwards. This post-challenge phase is specifically designed to bridge the gap between academic/start-ups innovation and real-world deployment.

Build the Future of India's Drone Ecosystem

If you have an idea that can redefine autonomous flight, intelligent sensing, aerial mobility, infrastructure automation, and the cybersecurity of drones, this is your opportunity.

Design. Build. Test. Demonstrate.

Join the **PUSHPAK Grand Challenge 2026** and help create the next generation of indigenous drone technologies for India and the world.

Objective:

The objective of the Grand Challenge is to support the detailed, **CAE-supported design of an indigenous cycloidal rotor module for drone applications.**

The challenge will focus on developing technically sound cyclorotor designs through kinematic, aerodynamic, structural, and integration-level analysis. The end goal of the Grand Challenge is not immediate prototype fabrication, but the identification of strong, build-ready cyclorotor designs.

Challenge Question:

Cycloidal rotors offer a promising alternative to conventional drone propulsion systems by enabling rapid thrust vectoring, improved maneuverability, and enhanced control authority. Unlike conventional propellers, cyclorotors can vary the direction of generated thrust through cyclic blade-pitch variation. This capability is useful for drones operating in confined spaces, near structures, in gusty environments, and in missions requiring precise attitude and orientation control.

However, cyclorotor-based drone technology is not widely available in India. Indigenous development of this technology requires focused work on blade-pitch kinematics, rotor geometry, aerodynamic performance, structural design, lightweight mechanisms, power requirements, and manufacturability.

The problem is to develop a detailed engineering design of a cycloidal rotor module suitable for drone applications. The design should target at least 10 N thrust and a thrust-to-weight ratio greater than 2.5. Teams must support their design through CAD models, kinematic analysis, aerodynamic calculations or simulations, structural assessment, mass estimation, material selection, actuation strategy, and an implementation plan.

The expected submission is a detailed design package that demonstrates technical feasibility and readiness for prototype development. The final selected designs will be eligible for a subsequent build phase with support from IIT Bombay.

Format of Competition:

Stage 1: Problem Statement Announcement and Preliminary Design

The Grand Challenge will begin with the announcement of the problem statement. Teams will be given one month to prepare and submit a preliminary design.

The preliminary design submission should include:

1. Cyclorotor concept and configuration.
2. Preliminary rotor sizing.
3. Blade arrangement and pitch-control concept.
4. Estimated thrust and power requirement.
5. Estimated module weight and thrust-to-weight ratio.
6. Initial material and manufacturing approach.
7. Team capability and execution plan.

No financial support will be provided at this stage. The objective of this stage is to identify technically promising teams for detailed design support.

The organizing team will evaluate preliminary designs and **shortlist up to 15 teams**. These teams will receive **Rs. 1 lakh** each to develop a complete, CAE-supported detailed design.

Stage 2: Detailed Design Development

During The detailed design phase period, shortlisted teams will prepare a full design package including:

1. CAD model of the complete cyclorotor module.
2. Kinematic model of the blade-pitch mechanism.
3. Aerodynamic analysis or simulation for thrust prediction.
4. Structural analysis of blades, supports, frame, shaft, and critical linkages.
5. Motor, actuator, bearing, and controller selection.
6. Material selection and mass estimate.
7. Thrust-to-weight ratio estimate.
8. Manufacturability and assembly plan.
9. Bill of materials and cost estimate.
10. Risk assessment and mitigation plan.
11. Build and test plan for the next phase.

After submission of detailed design reports, the organizing team will evaluate the submissions. **Up to 10 teams** will be **shortlisted** for final presentation during IIT Bombay Techfest in December 2026.

Stage 3: Detailed Design Evaluation

During this stage, the evaluation committee may provide brief feedback to the shortlisted teams. Teams will be encouraged to improve their designs, simulations, documentation, and presentation before the final round. Travel or participation support may be provided to the shortlisted finalist teams, subject to the approved budget norms.

Final Presentation: Techfest Evaluation

The final round will be conducted during IIT Bombay Techfest in December 2026. Finalist teams will present their detailed design before the evaluation committee and respond to technical questions.

The final evaluation will be based on:

1. Detailed design report.
2. CAD and CAE-supported technical evidence.
3. Final presentation.
4. Viva and technical defence.
5. Feasibility of implementation in the next phase.

Up to 5 winning teams will be selected. These winners will receive awards or recognition grants and will be considered for further financial support, technical mentoring, and access to resources for prototype development and testing with IIT Bombay support.

Evaluation Criteria

Criterion	Weight
Feasibility of achieving 10 N thrust	15%
Feasibility of achieving thrust-to-weight ratio greater than 2.5	15%
Kinematic design of blade-pitch and thrust-vectoring mechanism	15%

Aerodynamic analysis or simulation quality	15%
Structural design and strength assessment	15%
Manufacturability, material selection, and cost realism	10%
CAD quality, integration readiness, and packaging	5%
Presentation, viva, and technical clarity	10%
Total	100%

The final score will be based on the detailed design report, CAE-supported evidence, final presentation, and viva during the Techfest evaluation.

The target design requirements are:

Parameter	Target Requirement
Thrust capacity	At least 10 N
Thrust-to-weight ratio	Greater than 2.5
Rotor type	Cycloidal rotor with blade-pitch variation
Thrust vectoring	Demonstrated through kinematic and performance analysis
Design support	CAD and CAE-supported design
Analysis requirement	Kinematic, aerodynamic, and structural simulations
Final outcome	Build-ready cyclorotor design suitable for future prototype development

The thrust-to-weight ratio shall be estimated based on the complete cyclorotor module, including rotor blades, frame, pitch mechanism, motor, actuator, and associated mounting hardware.

Team Specification and Eligibility:

Expected Target Audience (Upto 5 members per team)

The Grand Challenge is intended for teams capable of developing a technically rigorous cyclorotor design supported by CAD, simulation, and engineering analysis.

The expected target audience includes:

1. Academic institutions working in UAVs, rotorcraft, aerodynamics, robotics, controls, or mechanical design.
2. Faculty-led student teams with strong design, simulation, and prototyping capability.
3. Startups developing drone platforms, propulsion systems, robotics hardware, or UAV subsystems.
4. MSMEs with capability in precision fabrication, lightweight structures, electric propulsion, embedded systems, or drone components.
5. Academia–startup–industry consortia interested in developing advanced indigenous UAV technologies.

Preference may be given to teams with demonstrated capability in one or more of the following areas:

1. Rotor design and unsteady aerodynamics.
2. CAD modelling and mechanical design.
3. Kinematic analysis of mechanisms.
4. CFD, FEA, multibody dynamics, or other CAE tools.
5. Lightweight structures and material selection.
6. Motor, actuator, and control-system selection.
7. UAV subsystem integration and testing.

Important Eligibility Restriction

Project staff, research staff, consultants, interns, and other personnel directly engaged with the PUSHPAK Project, and Drone Centre, or the organizing/host institutions in connection with the Grand Challenge are NOT eligible to participate in the Grand Challenge.

This restriction applies whether such personnel participate individually or as part of a team.

Any team found to include an ineligible participant may be disqualified at any stage of the Grand Challenge, including after selection or announcement of results. The organizers reserve the right to verify the eligibility of participants at any stage and their decision in this regard shall be final.

Timeline and Key Dates:

Stage	Timeline(s)	Participation Brief	Prize / Reward / Support
Grand Challenge Announcement	22 August 2026	PUSHPAK Grand Challenge 2026 opens for registrations across India. Teams of up to 5 members may register and participate.	None
Stage 1: Preliminary Design Submission	Work Time: 22 August – 27 September 2026 Submission Deadline: On or Before 27 September 2026 Results Announcement: 02 October 2026	All registered teams are eligible. Submit a preliminary design of the cyclorotor module. (Upto 15 Teams shall qualify for next stage)	INR 1 Lakh per qualifying team after the announcement of results of Stage 1
Stage 2: Detailed Design Development	Work Time: 03 October – 02 December 2026 Submission Deadline: On or Before 02 December 2026 Results Announcement: 06 December 2026	Shortlisted teams develop and submit a complete, CAE-supported detailed design package for the cyclorotor module. (Upto 10 Teams shall qualify for next stage)	Domestic Travel & Accommodation support as per guidelines of IIT Bombay to be shared at a later stage for qualifying team

Stage 3: Grand Challenge Finale	16 – 18 December 2026	Finalist teams present and demonstrate their detailed design before the evaluation committee at IIT Bombay Techfest. (Upto 5 Winning Teams gets chance to enter Post-Grand Challenge Stage)	Zenith – INR 3 Lakh, Horizon – INR 2.5 Lakhs, Vector – INR 2 Lakhs, Ascent (x2) – INR 1.5 Lakh, for the winning team after the announcement of results at the TechFest
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Registration & Submission

- Participants must register for the PUSHPAK Grand Challenge on the official Techfest website: techfest.org > Competitions > PUSHPAK Grand Challenge > Register at www.techfest.org/competitions
- All submissions (Stage 1 and Stage 2) are to be made via email to [pushpak_gc2026@aero.iitb.ac.in] — no submission links are used; every stage is handled entirely through email correspondence.

Prize Money

Stage 1 (Preliminary Design Verification): INR 1 Lakh per qualifying team after the announcement of results of Stage 1.

Stage 2 (Technical Challenge): Domestic Travel & Accommodation support as per guidelines of IIT Bombay to be shared at a later stage for qualifying teams.

Stage 3 (Grand Challenge Finale):

Zenith: INR 3 Lakh

Horizon: INR 2.5 Lakhs

Vector: INR 2 Lakhs

Ascent (x2): INR 1.5 Lakh

for winning teams after the announcement of results at the TechFest.

Note: At the end of the challenge, up to 5 winning teams will be selected. These teams will be considered for further support, funding, mentoring, and access to resources for building and testing their cyclorotor designs in the next phase.

The process regarding the disbursement of prize money will be communicated at a later stage. Participating teams are requested to regularly check the official website and their registered email accounts for updates.

Disclaimer:

Organizer's Rights: The organizers reserve the right to modify, extend, postpone, suspend, or cancel the Grand Challenge, or any stage thereof, at their discretion. Any such changes will be communicated through the official communication channels of the Grand Challenge.

Eligibility & Participation: Participation in the Grand Challenge is subject to the eligibility criteria, rules, guidelines, and submission requirements specified by the organizers. Submission of an entry does not guarantee selection for any subsequent stage or award.

Evaluation: All submissions will be evaluated by the designated evaluation committee based on the criteria specified for the respective stage. The decision of the evaluation committee shall be final and binding, and no correspondence regarding the evaluation or ranking will be entertained.

Technical Responsibility: Participants are solely responsible for the accuracy, completeness, originality, technical feasibility, safety, and performance of their submissions. Selection or recognition of a submission by the organizers does not constitute certification, approval, or endorsement of the proposed technology for operational deployment.

Safety & Compliance: Participants shall ensure that their proposed solutions and demonstrations comply with all applicable laws, regulations, standards, aviation requirements, safety protocols, and permissions applicable in India. Participants shall be responsible for obtaining any permissions, approvals, or clearances required for testing or demonstration.

Intellectual Property: Participants shall ensure that their submissions do not infringe upon any third-party intellectual property rights. Participants retain ownership of their intellectual property unless otherwise specifically agreed under the terms of the Grand Challenge. Any use, disclosure, licensing, or transfer of intellectual property arising from the Grand Challenge shall be governed by the applicable terms and agreements communicated by the organizers.

Confidentiality: Participants should clearly identify any information submitted that they consider confidential or proprietary. Participants are responsible for ensuring that they do not disclose information belonging to third parties without appropriate authorization.

Prizes & Recognition: Prizes, awards, certificates, or other forms of recognition are subject to fulfillment of the applicable eligibility and evaluation requirements. The organizers reserve the right to withhold or modify an award where no submission meets the required standards.

Expenses: Unless expressly stated otherwise, participation in the Grand Challenge, including development, testing, travel, accommodation, equipment, and demonstration-related expenses, shall be borne by the participating teams.

Liability: The organizers, host institutions, funding agencies, partners, mentors, evaluators, and their respective personnel shall not be liable for any loss, damage, injury, delay, technical failure, or other consequence arising from participation in the Grand Challenge, except to the extent required by applicable law.

Force Majeure: The organizers shall not be responsible for any delay, modification, suspension, or cancellation resulting from circumstances beyond their reasonable control, including natural disasters, technical failures, regulatory changes, government directives, or other unforeseen events.

Acceptance of Terms: By registering for or participating in the Grand Challenge, participants acknowledge that they have read, understood, and agreed to abide by the applicable rules, guidelines, terms, and conditions of the Grand Challenge.